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AIUB, Machine Learning Notes

MACHINE LEARNING

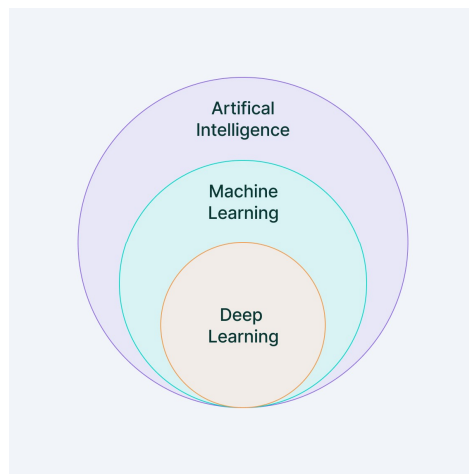
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INTRODUCTION TO MACHINE LEARNING

Introduction to Machine Learning

- ⇒ Machine Learning (ML) is a branch of Artificial Intelligence (AI) that focuses on building systems that can **learn from data, recognize patterns, and make decisions** with minimal human intervention.
- Instead of **being explicitly programmed**, ML models **improve automatically through experience**.
- **Example:** A spam filter learns from past emails marked as “spam” or “not spam” and automatically classifies new emails accordingly.



Applications of Machine Learning

- ⇒ ML is used in many real-world applications across industries:
- **Image Recognition**
- **Facial Recognition:** Used in smartphones and social media (e.g., Facebook photo tagging).
 - **Medical Imaging:** Identifies diseases or abnormalities in X-rays, MRIs, and CT scans.
 - **Object Detection:** Used in autonomous vehicles, security, and retail systems.
- **Speech and Voice Recognition**
- **Virtual Assistants:** Siri, Alexa, and Google Assistant use ML to process voice commands.
 - **Speech-to-Text:** Converts spoken words into text (e.g., Google Voice Typing).
 - **Voice Biometrics:** Used for secure identity verification based on voice patterns.

➤ **Natural Language Processing (NLP)**

- **Chatbots:** Provide customer support by understanding and responding to queries.
- **Language Translation:** Google Translate converts text or speech between languages.
- **Sentiment Analysis:** Helps companies analyze customer opinions on social media or reviews.

➤ **Recommendation Systems**

- **E-commerce:** Amazon suggests products based on previous purchases.
- **Social Media:** Facebook and Instagram suggest friends or content you might like.
- **Streaming Platforms:** Netflix and YouTube recommend shows or videos using viewing history.

➤ **Fraud Detection**

- **Banking:** Detects suspicious transactions (e.g., unusual credit card activity).
- **Insurance:** Flags fraudulent claim patterns.

➤ **Autonomous Vehicles**

- **Self-Driving Cars:** Use ML to detect pedestrians, traffic signs, and navigate roads safely.
- **Drones:** Use ML for route planning and obstacle avoidance.

➤ **Healthcare**

- **Disease Prediction:** Predicts risks for diseases like cancer or diabetes.
- **Personalized Medicine:** Creates tailored treatment plans based on medical data.
- **Drug Discovery:** Speeds up the process of identifying new drugs.

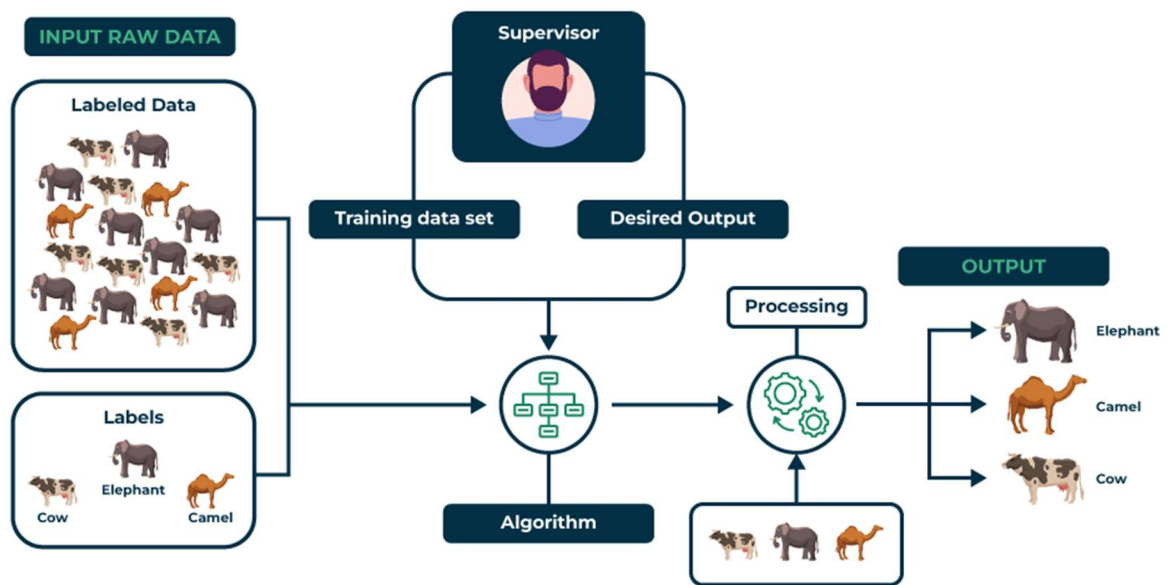
Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Semi-supervised Learning
4. Reinforcement Learning

Supervised Learning

⇒ The model learns from **labeled data**, each example has both input and correct output. The goal is to learn the mapping between inputs and outputs.

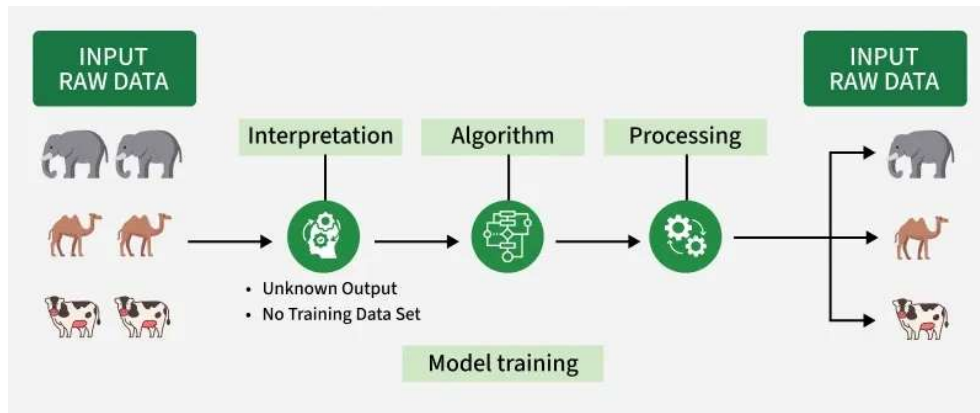
⇒ **Example:** Email spam detection (input: email content; output: spam or not spam).



Unsupervised Learning

The model learns from **unlabeled data** and discovers hidden structures or patterns.

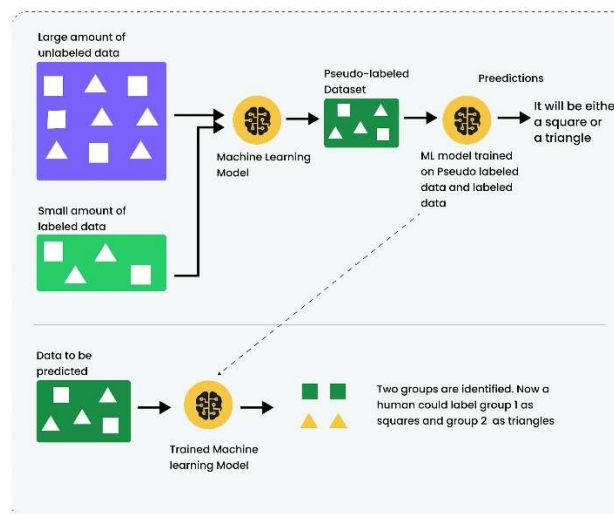
⇒ **Example:** Customer segmentation, grouping customers by similar purchasing habits without predefined labels.



Semi-supervised Learning

⇒ Uses a **small amount of labeled data** combined with a **large amount of unlabeled data** to improve accuracy.

⇒ **Example:** Image classification when only a few images are labeled but many are unlabeled.

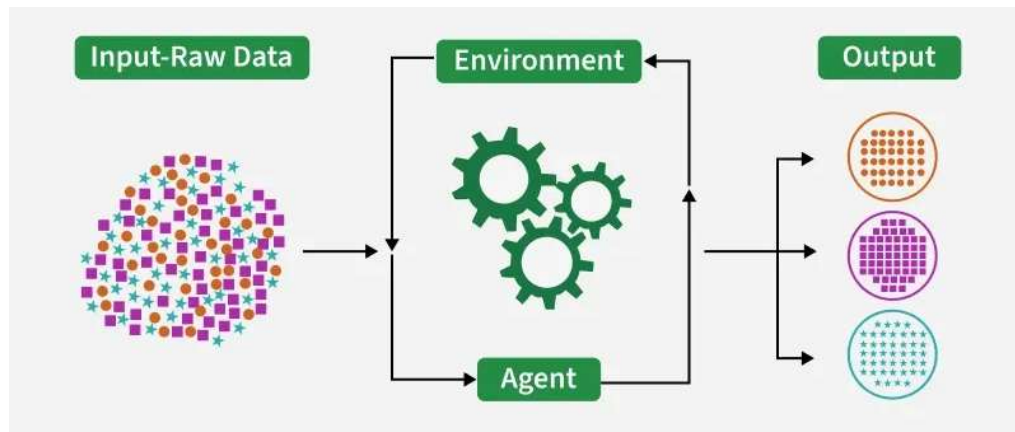


Reinforcement Learning

⇒ An **agent** interacts with an **environment** and learns through **rewards** or **penalties**. The agent's goal is to maximize total rewards over time.

⇒ **Example:** Training a robot to navigate a maze, rewarded when it reaches the goal.

⇒ **Another Example:** Ad placement on a webpage, an agent decides how many ads to show based on reward (revenue) feedback.



Difference Between Learning Types

Aspect	Supervised	Unsupervised	Reinforcement
Data Type	Labeled	Unlabeled	Interaction-based
Goal	Predict outcomes	Find patterns	Maximize rewards
Example	Spam filter	Customer groups	Game-playing robot

Key Concepts in Machine Learning

- ⇒ **Data & Features:** Raw data is the input; features are measurable attributes (e.g., height, weight). Labels are the expected outputs for supervised learning.
- ⇒ **Model & Algorithm:** The **model** represents what the system learns; **algorithms** are the methods used to train models (e.g., Linear Regression, Decision Trees).
- ⇒ **Training & Testing:** Data is split into **training data** (to teach the model) and **testing data** (to check performance).
- ⇒ **Loss Function:** Measures how far off predictions are from actual results (e.g., Mean Squared Error).
- ⇒ **Optimization:** Techniques like **Gradient Descent** adjust model parameters to minimize loss and improve accuracy.

Learning Process

➤ Steps:

1. Collect and prepare data
2. Train the model using an algorithm
3. Test the model with new data
4. Evaluate the results with performance metrics

➤ Evaluation Metrics:

- **Accuracy:** Overall correctness
- **Precision:** Correct positive predictions among predicted positives
- **Recall:** Ability to find all relevant cases
- **F1 Score:** Harmonic mean of precision and recall

Overfitting: Model memorizes data → **performs well on training** but **poorly on new data.**

Underfitting: Model is too simple → **fails to capture data patterns.**

Fundamental Algorithms Overview

1. **Linear Regression:** Predicts continuous values (e.g., house prices).
2. **Logistic Regression:** For binary classification (e.g., pass/fail, spam/not spam).
3. **Decision Trees:** Splits data into branches for decision-making.
4. **k-Nearest Neighbors (k-NN):** Classifies based on similarity to nearby samples.
5. **Neural Networks:** Mimic the human brain; used in deep learning (e.g., image or speech recognition).
6. **Support Vector Machines (SVM):** Finds the best dividing boundary between different classes.

Challenges in Machine Learning

- **Data Quality & Quantity:** Poor or limited data reduces model accuracy.
- **Feature Engineering:** Choosing the right input variables greatly affects model performance.
- **Interpretability:** Some models (like deep learning) act as “black boxes”, hard to explain decisions.
- **Scalability:** Handling massive datasets efficiently is a challenge.
- **Performance:** Balancing accuracy, speed, and memory usage for real-time systems.

REGRESSION

Supervised Learning Overview

⇒ In **supervised learning**, the model is trained using labeled data — meaning we know both the input and the desired output.

There are two main types of supervised learning tasks:

- **Regression:** Predicting a **numeric (continuous)** value.
- **Classification:** Predicting a **category or class**.

Examples:

- **Regression** → Predicting a car's price based on its mileage, brand, and age.
- **Classification** → Spam detection (classifying emails as *spam* or *not spam*).

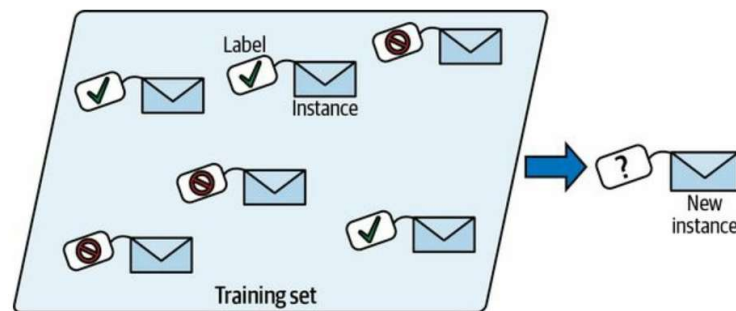


Fig: Supervised Learning: Classification

What is Regression?

⇒ Regression is a type of supervised learning where the goal is to **predict continuous output values** (e.g., temperature, price, or age) based on one or more input variables.

⇒ **Example:** Predicting the price of a house based on features like size, location, and number of rooms.

→ Objective:

To find the **best-fit line (or curve)** that represents the relationship between **input variables (X)** and the **output variable (Y)**.

This line should minimize the difference (error) between predicted and actual values.

Types of Regression

1. Simple Linear Regression
2. Multiple Linear Regression
3. Polynomial Linear Regression
4. Logistic Regression

Simple Linear Regression

⇒ Simple Linear Regression models the relationship between **one independent variable (X)** and **one dependent variable (Y)** using a straight line.

⇒ **Equation:**

$$y = b_0 + b_1x$$

Where:

- y → Predicted output (dependent variable)
- x → Input variable (independent variable)
- b_1 → Slope of the line (rate of change)
- b_0 → Intercept (value of y when $x = 0$)

➤ **Example:** Predicting a student's exam score (Y) based on hours studied (X):

$$y = 50 + 5x$$

- $b_1 = 5$ → Each extra hour of study increases score by 5 points.
- $b_0 = 50$ → Even with 0 study hours, expected score = 50.

If $x = 4$ hours:

$$y = 50 + 5 \times 4 = 70$$

So, the predicted score is **70**.

Real-world uses:

- Predicting sales based on advertising spend.
- Predicting temperature based on time of day.

Multiple Linear Regression

⇒ When there are **two or more independent variables**, the model is called *Multiple Linear Regression*.

⇒ **Equation:**

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$$

Where:

- y → Predicted output (dependent variable)
- x → Input variable (independent variable)
- $b_1, b_2, b_3, \dots$ → Slope of the line (rate of change)
- b_0 → Intercept (value of y when $x = 0$)

➤ **Example Scenario:** Predicting a person's monthly salary (Y) using:

- x_1 : Years of education
- x_2 : Years of work experience

$$y = 2000 + 1500x_1 + 800x_2$$

Interpretation:

- **Intercept (2000):** Base salary for zero education and experience.
- **$b_1 = 1500$:** Each additional year of education increases salary by \$1500.
- **$b_2 = 800$:** Each additional year of experience increases salary by \$800.

Prediction Example:

For 16 years of education and 5 years of experience:

$$y = 2000 + 1500 \times 16 + 800 \times 5 = 30,000$$

Predicted salary = **\$30,000**.

Real-world uses:

- Predicting house prices using multiple features (area, location, number of rooms).
- Forecasting revenue using advertising, market conditions, and product pricing.

Polynomial Linear Regression

⇒ Polynomial Regression models the relationship between X and Y as an **nth-degree polynomial**. Although the curve is non-linear, the model is *linear in coefficients*.

⇒ **Equation:**

$$y = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots + b_nx^{n-1}$$

Where:

- y → Predicted output (dependent variable)
- x → Input variable (independent variable)
- $b_1, b_2, b_3, \dots$ → Slope of the line (rate of change)
- b_0 → Intercept (value of y when $x = 0$)

Use Case:

- When data shows a **curved relationship** instead of a straight line.

Real-world uses:

- Predicting population growth or stock trends where change is not constant.

Logistic Regression (Overview)

⇒ Despite its name, **Logistic Regression** is used for **classification**, not regression. It predicts **categorical outcomes** (e.g., Yes/No, 0/1) using a **sigmoid function** to map values between 0 and 1.

⇒ **Equation:**

$$y = \frac{1}{1 + e^{-(b_0 + b_1x)}}$$

Example: Predicting whether an email is spam (1) or not spam (0).

Error and Cost Function in Linear Regression

What is an Error in Regression?

⇒ When a regression model makes predictions, the **error** (also called **residual**) is the **difference between the actual output and the predicted output**.

$$\text{Error (Residual)} = \text{Actual Value} - \text{Predicted Value}$$

In other words:

- If the prediction is **too high**, the error is **negative**.
- If the prediction is **too low**, the error is **positive**.

Example:

Actual (Y)	Predicted ($\hat{Y}$)	Error (Y - $\hat{Y}$)
50	48	+2
60	63	-3
70	68	+2

So, the model makes small mistakes — these differences are what we call **errors**.

Why Do We Need a Cost Function?

⇒ A **cost function** measures how well (or poorly) a regression model fits the data. It calculates the **overall error** of all predictions in the dataset.

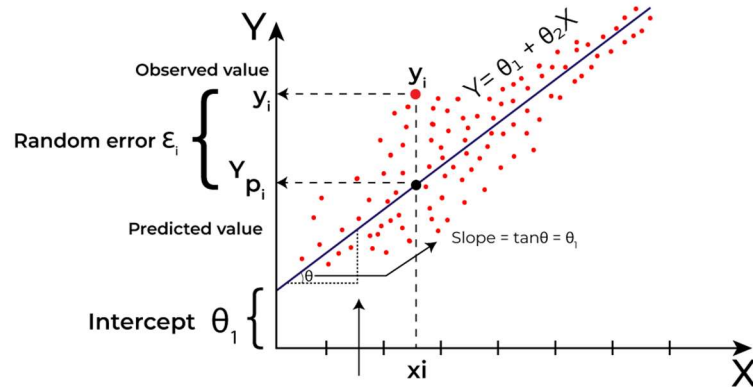
The goal of training a regression model is to **minimize the cost function** — meaning, we want to make predictions as close as possible to the real values.

The Concept of the Best-Fit Line

⇒ The **best-fit line** minimizes the **error** between actual and predicted values.

Goal: Minimize the **sum of squared errors (SSE)** using optimization techniques like **Gradient Descent**.

Why it matters: A well-fit line accurately represents the relationship between variables, improving prediction accuracy.



$$\hat{y}_i = \theta_1 + \theta_2 x_i$$

Mean Squared Error (MSE) – The Most Common Cost Function

⇒ The **Mean Squared Error (MSE)** is the most widely used cost function in regression.

$$MSE/J = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

Where:

- **n** = number of data points
- **y_i** = actual value
- **$\hat{y}_i$** = predicted value

Example:

Actual (Y)	Predicted ($\hat{Y}$)	Error (Y - $\hat{Y}$)	Error ²
4	5	-1	1
3	2.5	0.5	0.25
5	4	1	1
6	5.5	0.5	0.25

$$MSE = \frac{1 + 0.25 + 1 + 0.25}{4} = \frac{2.5}{4} = 0.625$$

So, the **mean squared error = 0.625**. A smaller MSE means a better-fitting regression model.

Root Mean Squared Error (RMSE)

⇒ RMSE is simply the **square root of MSE**, which brings the error value back to the same unit as the output variable.

$$RMSE = \sqrt{MSE}$$

Example: If $MSE = 0.625 \rightarrow$

$$RMSE = \sqrt{0.625} = 0.79$$

So, the average prediction error is around **0.79 units**.

Mean Absolute Error (MAE)

⇒ MAE is another way to measure error — instead of squaring, it takes the **absolute difference** between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

Example:

Actual (Y)	Predicted ($\hat{Y}$)	$ Y - \hat{Y} $
4	5	1
3	2.5	0.5
5	4	1
6	5.5	0.5

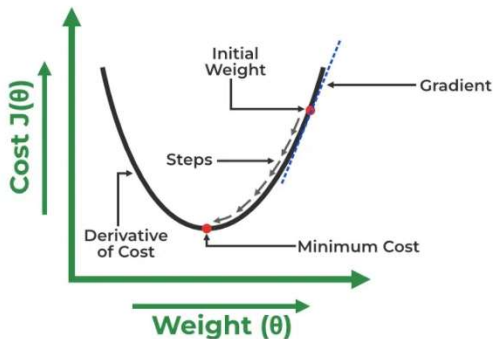
$$MAE = \frac{1 + 0.5 + 1 + 0.5}{4} = \frac{3}{4} = 0.75$$

So, On average, the model's prediction is off by **0.75 units**.

Gradient Descent for Linear Regression

⇒ **Gradient Descent** is an **optimization algorithm** used to **minimize the cost function** in machine learning models — especially in **linear regression**.

The main goal of linear regression is to find the **best-fit line**:



That line should make predictions that are as close as possible to the actual data points. To achieve this, we adjust the parameters (θ_2 = slope, θ_1 = intercept) until the **cost function (error)** is minimized.

Gradient Descent helps us **find those optimal parameter values** step by step.

The Gradient Descent Update Rule

⇒ We update both parameters iteratively:

$$\theta_1 = \theta_1 - \alpha \frac{\partial J}{\partial \theta_1}$$

$$\theta_2 = \theta_2 - \alpha \frac{\partial J}{\partial \theta_2}$$

Where:

- α = **Learning rate** (controls how big each update step is)
- $\frac{\partial J}{\partial \theta_1}, \frac{\partial J}{\partial \theta_2}$ = gradients of the cost function

Derivation of the Gradients

⇒ Let's differentiate the cost function with respect to each parameter.

Derivative w.r.t. θ_1	Derivative w.r.t. θ_2
$J'_{\theta_1} = \frac{\partial}{\partial \theta_1} \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) \frac{\partial}{\partial \theta_1} (\hat{y}_i - y_i) \because \hat{y}_i = \theta_1 + \theta_2 x_i$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) \frac{\partial}{\partial \theta_1} (\theta_1 + \theta_2 x_i - y_i)$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) (1 + 0 - 0)$ $J'_{\theta_1} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \dots \dots \dots (i)$	$J'_{\theta_2} = \frac{\partial}{\partial \theta_2} \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) \frac{\partial}{\partial \theta_2} (\hat{y}_i - y_i)$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) \frac{\partial}{\partial \theta_2} (\theta_1 + \theta_2 x_i - y_i)$ $= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i) (0 + x_i - 0)$ $J'_{\theta_2} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i \dots \dots \dots (ii)$
Final Gradient Descent Update Formulas:	
$\theta_1 = \theta_1 - \alpha \times \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$	$\theta_2 = \theta_2 - \alpha \times \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$
These formulas iteratively adjust θ_1 and θ_2 to reduce the error and reach the minimum point of the cost function.	

✂ **Problem 1:** We are given the following dataset showing the relationship between Age (X) and Salary (Y):

Age (X)	Salary (Y)
30	800
37	950
25	600
43	1050
50	1200
29	740
46	1100

A linear regression model is used to predict salary from age using the hypothesis function:

$$\hat{y}_i = \theta_1 + \theta_2 x_i$$

$$\theta_1=300, \theta_2=10, \alpha=0.0001$$

Perform **one iteration of the Gradient Descent algorithm**, and compute:

1. The initial cost J (Mean Squared Error).
2. The gradients J'_{θ_1} and J'_{θ_2}
3. The updated values of θ_1 & θ_2
4. Write the updated regression equation.
5. The new cost J_{new} after parameter updates.

Solution

(1)

Is given,

$$\text{Hypothesis Function: } \hat{y}_i = \theta_1 + \theta_2 x_i$$

$$\text{Substitute given values: } \hat{y}_i = 300 + 10x_i$$

$$n = 7$$

We know,

$$J = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \frac{1}{7} \sum_{i=1}^7 (\hat{y}_i - y_i)^2 \dots \dots \dots (i)$$

$$\sum_{i=1}^7 (\hat{y}_i - y_i)^2 = (300 + 10 \times 30 - 800)^2 + (300 + 10 \times 37 - 950)^2 + (300 + 10 \times 25 - 600)^2 + (300 + 10 \times 43 - 1050)^2 \\ + (300 + 10 \times 50 - 1200)^2 + (300 + 10 \times 29 - 740)^2 + (300 + 10 \times 46 - 1100)^2 = \mathbf{521399.998}$$

Put the value in equation 1:

$$J = \frac{1}{7} \sum_{i=1}^7 (\hat{y}_i - y_i)^2 = \frac{1}{7} (521399.998) = \mathbf{74485.714}$$

(2)

Gadients J'_{θ_1}	Gadients J'_{θ_2}
$J'_{\theta_1} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$ $J'_{\theta_1} = \frac{2}{n} \sum_{i=1}^n (300 + 10x_i - y_i)$ $= \mathbf{-497.1412}$	$J'_{\theta_2} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$ $J'_{\theta_2} = \frac{2}{n} \sum_{i=1}^n (300 + 10x_i - y_i) x_i$ $= \mathbf{-20388.57412}$

(3)

NEW θ_1	NEW θ_2
$\theta_1 = \theta_1 - \alpha \times J'_{\theta_1}$ $= 300 - 0.0001 \times -497.1412$ $= \mathbf{300.0497}$	$\theta_2 = \theta_2 - \alpha \times J'_{\theta_2}$ $= 10 - 0.0001 \times -20388.57412$ $= \mathbf{12.03}$

(4)

Is given,

Hypothesis Function: $\hat{y}_i = \theta_1 + \theta_2 x_i$

Updated equation be: $\hat{y}_i = 300.0497 + 12.03x_i$

(5)

We know,

$$J = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \frac{1}{7} \sum_{i=1}^7 (\hat{y}_i - y_i)^2 = 39084.8289$$

Linear Regression in Matrix Form

⇒ In machine learning, especially when working with multiple variables, it's more convenient to represent linear regression using **matrix notation**.

This makes computation simpler and more efficient — especially when using tools like **NumPy**, **MATLAB**, or **TensorFlow**.

⇒ **Equation:**

$$Y = Xw$$

Where:

- $Y \rightarrow$ Output (dependent variable)
- $X \rightarrow$ Input features (independent variables)
- $w \rightarrow$ Weight vector (parameters, including intercept)

If we include an intercept term θ_0 (bias), we modify X as:

$$Y = Xw = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} \theta_0 + \theta_1 x_1 \\ \theta_0 + \theta_1 x_2 \\ \vdots \\ \theta_0 + \theta_1 x_n \end{bmatrix}$$

So, each row represents one training example.

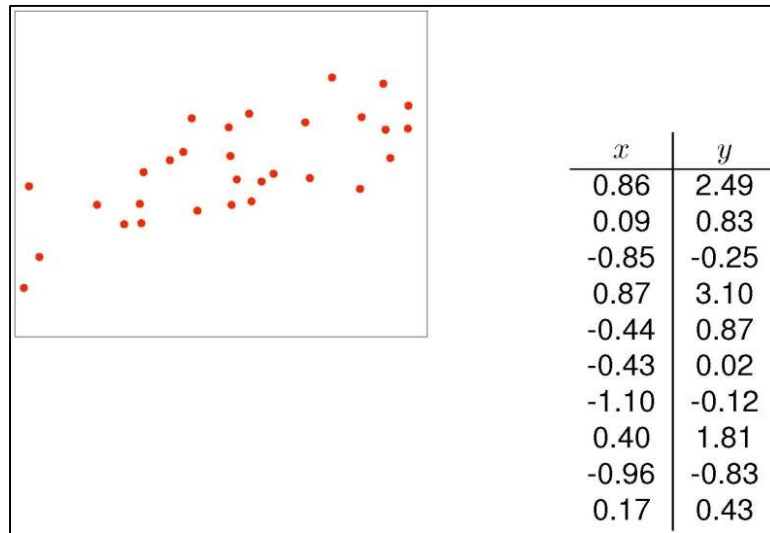
➤ **Example Dataset:** Let's take a small example with **two data points**:

X (Hours Studied)	Y (Score)
1	2
2	3

Here, we are trying to find the best-fit line $Y = \theta_0 + \theta_1 X$

Step1: Representing Data in Matrix Form	Step2: To find the best-fit parameters (weights), we use the Normal Equation:
$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad Y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ The first column of 1's in X corresponds to the intercept term θ_0	$w = (X^T X)^{-1} X^T Y$ Where: • X^T = Transpose of X • $(X^T X)^{-1}$ = Inverse of the product $X^T X$ • w = Parameter vector $[\theta_0, \theta_1]^T$
Step3: Step-by-Step Calculation	
Step3.1: Compute $X^T X$	Step3.2: Compute $X^T Y$
$X^T X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$	$X^T Y = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$
Step3.3: Compute $(X^T X)^{-1}$	Step3.3: Compute $w = (X^T X)^{-1} X^T Y$
$(X^T X)^{-1} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$	$w = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} \times \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Step4: Final Model Equation for best fit line	
$Y = 1 + 1X$	

Problem 2: You are given the following dataset showing the relationship between hours studied (X) and exam scores (Y):



- a) Using **Linear Regression in Matrix Form**, find the **best-fit line equation** $Y = \theta_0 + \theta_1 X$
- b) by computing all steps manually: $\theta = (X^T X)^{-1} X^T Y$, Then, **predict the score** when a student studies for **4 hours**.

Solutions

(a)

Step1: Representing Data in Matrix Form	Step2: is given Normal Equation:
$X = \begin{bmatrix} 1 & 0.86 \\ 1 & 0.09 \\ 1 & -0.85 \\ 1 & 0.87 \\ 1 & -0.44 \\ 1 & -0.43 \\ 1 & -1.10 \\ 1 & 0.40 \\ 1 & -0.96 \\ 1 & 0.17 \end{bmatrix}, \quad Y = \begin{bmatrix} 2.49 \\ 0.83 \\ -0.25 \\ 3.10 \\ 0.87 \\ 0.02 \\ -0.12 \\ 1.81 \\ -0.83 \\ 0.43 \end{bmatrix}$	$\theta = (X^T X)^{-1} X^T Y$ Where: X^T = Transpose of X $(X^T X)^{-1}$ = Inverse of the product $X^T X$ θ = Parameter vector $[\theta_0, \theta_1]^T$

Step3: Step-by-Step Calculation

Step3.1: Compute $X^T X$

$$X^T X = \begin{bmatrix} 1 & 0.86 \\ 1 & 0.09 \\ 1 & -0.85 \\ 1 & 0.87 \\ 1 & -0.44 \\ 1 & -0.43 \\ 1 & -1.10 \\ 1 & 0.40 \\ 1 & -0.96 \\ 1 & 0.17 \end{bmatrix}^T \times \begin{bmatrix} 1 & 0.86 \\ 1 & 0.09 \\ 1 & -0.85 \\ 1 & 0.87 \\ 1 & -0.44 \\ 1 & -0.43 \\ 1 & -1.10 \\ 1 & 0.40 \\ 1 & -0.96 \\ 1 & 0.17 \end{bmatrix} = \begin{bmatrix} 4.95 & -1.39 \\ -1.39 & 10 \end{bmatrix}$$

Step3.2: Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 0.86 \\ 1 & 0.09 \\ 1 & -0.85 \\ 1 & 0.87 \\ 1 & -0.44 \\ 1 & -0.43 \\ 1 & -1.10 \\ 1 & 0.40 \\ 1 & -0.96 \\ 1 & 0.17 \end{bmatrix}^T \times \begin{bmatrix} 2.49 \\ 0.83 \\ -0.25 \\ 3.10 \\ 0.87 \\ 0.02 \\ -0.12 \\ 1.81 \\ -0.83 \\ 0.43 \end{bmatrix} = \begin{bmatrix} 6.49 \\ 8.34 \end{bmatrix}$$

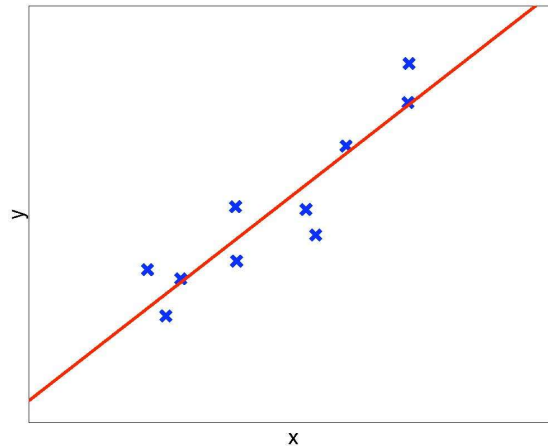
Step3.3: Compute $\theta = (X^T X)^{-1} X^T Y$

$$\theta = \begin{bmatrix} 4.95 & -1.39 \\ -1.39 & 10 \end{bmatrix}^{-1} \times \begin{bmatrix} 6.49 \\ 8.34 \end{bmatrix} = \begin{bmatrix} 1.60 \\ 1.05 \end{bmatrix}$$

Step4: Final Model Equation for best fit line

$$Y = 1.60 + 1.05X$$

Step5: figure



(b)

Predict the Score when X = 4:

$$Y = 1.60 + 1.05X$$

$$Y = 1.60 + 1.05 \times 4 = 5.800$$

Logistic Regression

- Used to **predict the probability** that something belongs to a certain class.
- Example: Is an email **spam (1)** or **not spam (0)**?

Binary Classification:

- Predicts **two classes only**:
 - Positive class $\rightarrow 1$
 - Negative class $\rightarrow 0$
- Uses a **threshold** (usually 0.5) to decide the class:
 - Probability $\geq 0.5 \rightarrow$ predict **1**
 - Probability $< 0.5 \rightarrow$ predict **0**

How it works:

1. Like linear regression, it computes a **weighted sum of inputs + bias**:

$$\hat{y}_i = \theta_1 + \theta_2 x_i \text{ becomes } \hat{y}_i = \sigma(\theta_1 + \theta_2 x_i)$$

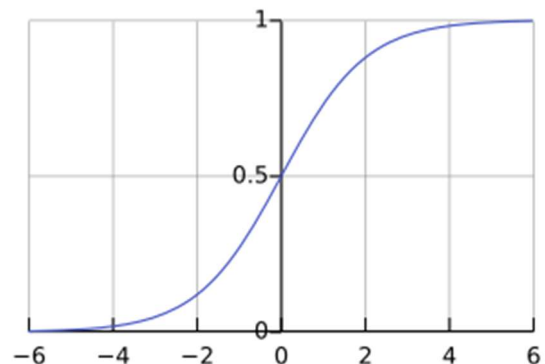
2. Then, instead of giving the sum directly, it passes it through the **sigmoid (logistic) function**:

$$\sigma(t) = \frac{1}{1+e^{-t}}$$

3. The output is a **probability between 0 and 1**.

Key points about the sigmoid function:

- $\sigma(t) < 0.5 \rightarrow t < 0 \rightarrow$ predicts **0**
- $\sigma(t) \geq 0.5 \rightarrow t \geq 0 \rightarrow$ predicts **1**



K-NEAREST NEIGHBORS (KNN)

K-Nearest Neighbors (KNN)

⇒ KNN is a **simple classification algorithm** that works by **finding the nearest data points** and choosing the most common class among them.

Key Features

Term	Meaning
Non-parametric	Does not assume any data pattern (flexible)
Instance-based / Lazy learner	Does not build a model; stores training data
Memory-based	Needs the data stored for prediction

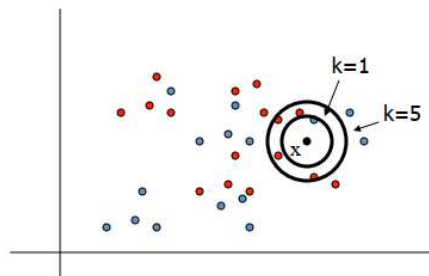
How KNN Works (Step-by-Step)

1. **Choose the value of K** (number of neighbors, like $K=3$ or $K=5$)
2. **Calculate distance** between new point & all training points
(Common distances: **Euclidean, Manhattan**)
3. **Pick K nearest neighbors**
4. **Majority Vote** → The class most common among the K neighbors → result

➤ **Example Dataset:** Suppose $K = 3$ and among the **3 closest neighbors**:

Neighbor	Class
1	Cat
2	Dog
3	Cat

Result = **Cat** (because Cat appears more)



common values for k: 3, 5

Distance Formula (Euclidean)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

🔗 **Problem 1:** You are given a dataset of fruits based on two features:

- **Weight (grams)**
- **Sweetness level (1–10)**

Fruit	Weight	Sweetness
Apple	180	7
Apple	160	6
Orange	200	4
Orange	220	3
Banana	120	8
Banana	130	9

A new fruit arrives with: **Weight = 170 g, Sweetness = 7**

a) Use **KNN with K = 3** to classify which fruit it is.

Solutions:

⇒ Compute Euclidean Distance:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Fruit	Coordinates	Distance
Apple (180,7)	(180,7)	$\sqrt{(170 - 180)^2 + (7 - 7)^2} = 10.0$
Apple (160,6)	(160,6)	$\sqrt{(170 - 160)^2 + (7 - 6)^2} = 10.05$
Orange (200,4)	(200,4)	$\sqrt{(170 - 200)^2 + (7 - 4)^2} = 30.15$
Orange (220,3)	(220,3)	$\sqrt{(170 - 220)^2 + (7 - 3)^2} = 50.16$
Banana (120,8)	(120,8)	$\sqrt{(170 - 120)^2 + (7 - 8)^2} = 50.01$
Banana (130,9)	(130,9)	$\sqrt{(170 - 130)^2 + (7 - 9)^2} = 40.05$

Pick 3 Nearest Neighbors:

Fruit	Distance
Apple (180,7)	10.00
Apple (160,6)	10.05
Orange (200,4)	30.15

Final Prediction: Predicted fruit = Apple

✂ **Problem 2:** You are given a dataset of fruits based on two features:

- height (cm)
- Weight (kg)

Height (cm)	Weight (kg)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal

A new people arrives with: **Height = 170 cm, Weight = 57**

a) Using the above dataset and **KNN algorithm with K = 3**, classify whether the person is **Underweight** or **Normal**.

Solutions:

⇒ Compute Euclidean Distance:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Coordinates	Distance
(167, 51)	$\sqrt{(170 - 167)^2 + (57 - 51)^2} = 6.70$
(182, 62)	$\sqrt{(170 - 182)^2 + (57 - 62)^2} = 13.0$
(176, 69)	$\sqrt{(170 - 176)^2 + (57 - 69)^2} = 13.4$
(173, 64)	$\sqrt{(170 - 173)^2 + (57 - 64)^2} = 7.6$
(172, 65)	$\sqrt{(170 - 172)^2 + (57 - 65)^2} = 8.2$
(174, 56)	$\sqrt{(170 - 174)^2 + (57 - 56)^2} = 4.1$
(169, 58)	$\sqrt{(170 - 169)^2 + (57 - 58)^2} = 1.4$
(173, 57)	$\sqrt{(170 - 173)^2 + (57 - 57)^2} = 3$
(170, 55)	$\sqrt{(170 - 170)^2 + (57 - 55)^2} = 2$

Arrange the distances in ascending order:

Height	Weight	Class	Distance
169	58	Normal	1.41
170	55	Normal	2.00
173	57	Normal	3.00
174	56	Underweight	4.12
167	51	Underweight	6.71
173	64	Normal	7.62
172	65	Normal	8.25
182	62	Normal	13.00
176	69	Normal	13.42

Majority Vote

Class	Count
Normal	3
Underweight	0

Final Prediction: Predicted Class = Normal

✂ **Problem 3:** A movie streaming platform, **CineMatch**, is developing a recommendation system that classifies new movies into genres based on their characteristics. The data science team decides to use the **K-Nearest Neighbors (KNN)** algorithm for genre classification.

They collect data from six previously released movies, each described by:

- **Length of the movie (in minutes)**
- **Number of action scenes**
- **Number of romantic scenes**

Each movie also has a known **Genre**.

Movie	Length (min)	Action Scenes	Romantic Scenes	Genre
M1	150	12	2	Action
M2	140	10	1	Action
M3	120	3	10	Romantic
M4	110	2	9	Romantic
M5	130	5	4	Comedy
M6	125	4	5	Comedy

Now, a **new movie**, *M7 — “Love & Fury”*, has just been added to the platform. It has the following attributes:

- Length = **135 minutes**
- Action Scenes = **6**
- Romantic Scenes = **3**

a) Based on the given data, use the **K-Nearest Neighbors (KNN)** algorithm with **K = 5** to determine the most suitable **genre** for the new movie “*Love & Fury*”. Show your reasoning or steps briefly.

Solutions:

⇒ Compute Euclidean Distance:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Coordinates	Distance
(150, 12, 2)	$\sqrt{(135 - 150)^2 + (6 - 12)^2 + (3 - 2)^2} = 16.19$
(140, 10, 1)	$\sqrt{(135 - 140)^2 + (6 - 10)^2 + (3 - 1)^2} = 6.71$

(120, 3, 10)	$\sqrt{(135 - 120)^2 + (6 - 3)^2 + (3 - 10)^2} = 16.82$
(110, 2, 9)	$\sqrt{(135 - 110)^2 + (6 - 2)^2 + (3 - 9)^2} = 26.02$
(130, 5, 4)	$\sqrt{(135 - 130)^2 + (6 - 5)^2 + (3 - 4)^2} = 5.20$
(125, 4, 5)	$\sqrt{(135 - 125)^2 + (6 - 4)^2 + (3 - 5)^2} = 10.39$

Arrange the distances in ascending order:

Length (min)	Action Scenes	Romantic Scenes	Genre	Distance
130	5	4	Comedy	5.20
140	10	1	Action	6.71
125	4	5	Comedy	10.39
150	12	2	Action	16.19
120	3	10	Romantic	16.82
110	2	9	Romantic	26.02

Majority Vote

Class	Count
Comedy	2
Action	2
Romantic	1

Handle the tie (Comedy vs Action): Use the smaller K (e.g., K=3) as a tiebreaker

Class	Count
Comedy	2
Action	1
Romantic	0

Final Prediction: Predicted Genre = Comedy

Advantages of KNN

Benefit	Why
Simple & easy to understand	No complex math
No training time	Only stores data
Can be used for classification & regression	Flexible
No assumptions on data	Works for many data types

Disadvantages of KNN

Issue	Explanation
Slow for large datasets	Calculates distance to every point
Needs memory to store data	All data must be saved
Sensitive to noise/outliers	Wrong results if data is messy
Feature scaling required	Variables with high values dominate distance

When to Use KNN

➤ Use KNN when:

- Dataset is **small to medium-sized**
- Data is **labeled**
- No strong assumption about data pattern

➤ Don't use when:

- Dataset is **very large**
- Real-time fast prediction is required

Important Concepts

Term	Meaning
Lazy learner	No training phase
Majority vote	Most neighbors' class wins
Distance metric	Way of measuring closeness

DECISION TREE

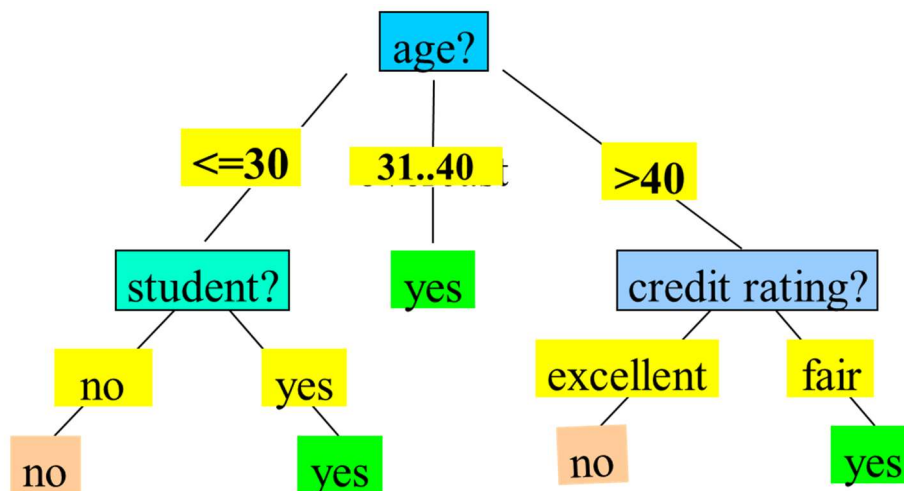
Introduction to Decision Trees

- ⇒ A **Decision Tree** is a **supervised machine learning algorithm** used for **classification** and **regression** tasks.
- ⇒ It works by splitting data into branches based on certain decision rules, forming a structure that looks like a **tree**.
 - Each **node** represents a **decision** based on a feature.
 - Each **branch** represents an **outcome** of that decision.
 - Each **leaf node** represents a **final prediction or class label**.
 - The **root node** is the starting point (the top of the tree).

Example: If you want to predict whether someone will buy a computer based on features like *age*, *income*, *student status*, and *credit rating*,

How Decision Trees Work

- ⇒ Decision Trees follow a **Top-Down Recursive Divide-and-Conquer** approach:
 1. Start with all data at the root.
 2. Select the **best feature** to split the data.
 3. Split the dataset into subsets based on feature values.
 4. Repeat the process for each subset.
 5. Stop when:
 - All samples in a node belong to the same class.
 - No more features to split.
 - A stopping condition is met.



ID3 Algorithm (Iterative Dichotomiser 3)

⇒ ID3 is one of the earliest and most famous algorithms for building **classification** trees.

❖ Purpose of ID3:

- Build a **decision tree** that classifies data with the **least uncertainty**.
- Selects attributes that provide the **highest information gain**.
- Splits the dataset recursively until:
 - All instances are classified, or
 - No attributes remain.

Steps in the ID3 Algorithm

Step 1: Calculate Entropy	Step 2: Compute Information Gain
⇒ Measures impurity or randomness.	⇒ Measures reduction in entropy after splitting the dataset.
$Entropy(D) = - \sum_{i=1}^n p_i \log_2(p_i)$ Where, p_i = proportion of class i, $p_i = \frac{n_i}{N}$ N = total sample n_i = Samples belonging to class i	$IG(D, A) = Entropy(D) - \sum \frac{ D_v }{ D } \times Entropy(D_v)$ Where, D = Original dataset. A = Attribute being evaluated. D_v = Subset of D where attribute A takes value v.
Step 3: Select Attribute with Highest Information Gain	
⇒ The attribute with the highest information gain becomes the decision node .	
Step 4: Split Dataset	
⇒ Partition data by attribute values.	
Step 5: Recursion:	
⇒ Repeat until: ⇒ All data in subset belongs to the same class ⇒ No attributes left ⇒ Dataset becomes empty	

🔗 **Problem1:** Consider the following dataset of 14 customers with attributes **age**, **income**, **student**, **credit_rating**, and the target class **buys_computer**.

Dataset:

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

❖ **Your Tasks is:**

1. Calculate the entropy of the entire dataset for the target attribute *buys_computer*.
2. Compute the **Information Gain (IG)** for each attribute:
 - age
 - income
 - student
 - credit_rating
3. Determine the best attribute for the root node based on highest Information Gain.
4. Using your root node, **perform the next split** on the corresponding subsets and show the resulting smaller decision tree.
5. Finally, use the constructed decision tree to classify the following new instance:

age = ≤ 30

income = low

student = yes

credit_rating = fair

Predict the value of the target attribute: **buys_computer?**.

(1)

Is given,

Totals: **14 datasets** $\rightarrow$ $\underbrace{9}_{\text{class 1}} \text{ yes}, \underbrace{5}_{\text{class 2}} \text{ no}.$

$$\therefore N = 14$$

$$\therefore n_1 = 9$$

$$\therefore n_2 = 5$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{9}{14} = 0.643$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{5}{14} = 0.357$$

now,

$$Entropy(D) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(D) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$Entropy(D) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$Entropy(D) = -0.643 \log_2(0.643) - 0.357 \log_2(0.357)$$

$$Entropy(D) = 0.940$$

$\therefore$ Entropy of the entire dataset for the target attribute **buys_computer**, $D = 0.940$

(2)

We Know,

$$IG(D, A) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)$$

❖ For Age (values: ≤ 30 , 31..40, > 40):

Is given, For Age ≤ 30 :

Totals for ≤ 30 : 5 datasets $\rightarrow$ $\underbrace{2 \text{ yes}}_{\text{class 1}}, \underbrace{3 \text{ no}}_{\text{class 2}}$.

$$\therefore N_{\leq 30} = 5$$

$$\therefore n_1 = 2$$

$$\therefore n_2 = 3$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{2}{5} = 0.400$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{3}{5} = 0.600$$

now,

$$\text{Entropy}(D_{v \leq 30}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \leq 30}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \leq 30}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v \leq 30}) = -0.400 \log_2(0.400) - 0.600 \log_2(0.600)$$

$$\text{Entropy}(D_{v \leq 30}) = 0.971$$

Is given, For Age 31 ... 40:

Totals for 31 ... 40 : 4 datasets $\rightarrow$ $\underbrace{4 \text{ yes}}_{\text{class 1}}, \underbrace{0 \text{ no}}_{\text{class 2}}$.

$$\therefore N_{31 \dots 40} = 4$$

$$\therefore n_1 = 4$$

$$\therefore n_2 = 0$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{4}{4} = 1.000$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{0}{4} = 0.000$$

now,

$$\text{Entropy}(D_{v_{31...40}}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v_{31...40}}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v_{31...40}}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v_{31...40}}) = -1.000 \log_2(1.000) - 0.000 \log_2(0.000)$$

$$\text{Entropy}(D_{v_{31...40}}) = 0.000 \text{ (pure)}$$

Is given, For Age > 40:

Totals for > 40 : 5 datasets → $\underbrace{3}_{\text{class 1}} \text{ yes}, \underbrace{2}_{\text{class 2}} \text{ no}.$

$$\therefore N_{>40} = 5$$

$$\therefore n_1 = 3$$

$$\therefore n_2 = 2$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{3}{5} = 0.600$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{2}{5} = 0.400$$

now,

$$\text{Entropy}(D_{v>40}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v>40}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v>40}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v>40}) = -0.600 \log_2(0.600) - 0.400 \log_2(0.400)$$

$$\text{Entropy}(D_{v>40}) = 0.971$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \leq 30}|}{|D|} \text{Entropy}(D_{v \leq 30}) + \frac{|D_{v 31 \dots 40}|}{|D|} \text{Entropy}(D_{v 31 \dots 40}) + \frac{|D_{v > 41}|}{|D|} \text{Entropy}(D_{v > 41})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|5|}{|14|} \times 0.971 + \frac{|4|}{|14|} \times 0.000 + \frac{|5|}{|14|} \times 0.971$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.694$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Age}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Age}) = 0.940 - 0.694 = 0.246$$

❖ **For Income (values: high, medium, low):**
Is given, For income, High:

Totals for **High** : 4 datasets $\rightarrow \underbrace{2 \text{ yes}}_{\text{class 1}}, \underbrace{2 \text{ no}}_{\text{class 2}}$.

$$\therefore N_{\text{high}} = 4$$

$$\therefore n_1 = 2$$

$$\therefore n_2 = 2$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{2}{4} = 0.500$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{2}{4} = 0.500$$

now,

$$\text{Entropy}(D_{v \text{ high}}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ high}}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ high}}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v \text{ high}}) = -0.500 \log_2(0.500) - 0.500 \log_2(0.500)$$

$$\text{Entropy}(D_{v \text{ high}}) = 1.000$$

Is given, For Income, medium:

Totals for **medium**: 6 datasets $\rightarrow \underbrace{4 \text{ yes}}_{\text{class 1}}, \underbrace{2 \text{ no}}_{\text{class 2}}$.

$$\therefore N_{\text{medium}} = 6$$

$$\therefore n_1 = 4$$

$$\therefore n_2 = 2$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{4}{6} = 0.667$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{2}{6} = 0.333$$

now,

$$\text{Entropy}(D_{v \text{ medium}}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ medium}}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ medium}}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v \text{ medium}}) = -0.667 \log_2(0.667) - 0.333 \log_2(0.333)$$

$$\text{Entropy}(D_{v \text{ medium}}) = 0.918$$

Is given, For Income, low.

Totals for **low**: 4 datasets $\rightarrow \underbrace{3}_{\text{class 1}} \text{ yes}, \underbrace{1}_{\text{class 2}} \text{ no}.$

$$\therefore N_{\text{low}} = 4$$

$$\therefore n_1 = 3$$

$$\therefore n_2 = 1$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{3}{4} = 0.750$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{1}{4} = 0.250$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

now,

$$\text{Entropy}(D_{v \text{ low}}) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ low}}) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$\text{Entropy}(D_{v \text{ low}}) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$\text{Entropy}(D_{v \text{ low}}) = -0.750 \log_2(0.750) - 0.250 \log_2(0.250)$$

$$\text{Entropy}(D_{v \text{ low}}) = 0.811$$

$$\sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ high}}|}{|D|} \text{Entropy}(D_{v \text{ high}}) + \frac{|D_{v \text{ medium}}|}{|D|} \text{Entropy}(D_{v \text{ medium}}) + \frac{|D_{v \text{ low}}|}{|D|} \text{Entropy}(D_{v \text{ low}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|4|}{|14|} \times 1.000 + \frac{|6|}{|14|} \times 0.918 + \frac{|4|}{|14|} \times 0.811$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.911$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Income}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Income}) = 0.940 - 0.911 = 0.029$$

❖ For Student (values: yes, no):

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

$$\text{Entropy}(D_{v \text{ no}}) = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right)$$

$$\text{Entropy}(D_{v \text{ no}}) = 0.985$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

$$\text{Entropy}(D_{v \text{ yes}}) = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right)$$

$$\text{Entropy}(D_{v \text{ yes}}) = 0.592$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ no}}|}{|D|} \text{Entropy}(D_{v \text{ no}}) + \frac{|D_{v \text{ yes}}|}{|D|} \text{Entropy}(D_{v \text{ yes}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|7|}{|14|} \times 0.985 + \frac{|7|}{|14|} \times 0.592$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.789$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Student}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Student}) = 0.940 - 0.789 = 0.152$$

❖ For credit_rating (values: fair, excellent):

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31 ... 40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31 ... 40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31 ... 40	medium	no	excellent	yes
31 ... 40	high	yes	fair	yes
> 40	medium	no	excellent	no

$\text{Entropy}(D_{v \text{ fair}}) = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right)$ $\text{Entropy}(D_{v \text{ fair}}) = 0.811$				
$\text{Entropy}(D_{v \text{ excellent}}) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right)$ $\text{Entropy}(D_{v \text{ excellent}}) = 1.000$				

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ fair}}|}{|D|} \text{Entropy}(D_{v \text{ fair}}) + \frac{|D_{v \text{ excellent}}|}{|D|} \text{Entropy}(D_{v \text{ excellent}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|8|}{|14|} \times 0.811 + \frac{|6|}{|14|} \times 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.892$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{credit_rating}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{credit_rating}) = 0.940 - 0.892 = 0.048$$

(3)

We have,

$$IG(D, \text{Age}) = 0.940 - 0.694 = 0.246$$

$$IG(D, \text{Income}) = 0.940 - 0.911 = 0.029$$

$$IG(D, \text{Student}) = 0.940 - 0.789 = 0.152$$

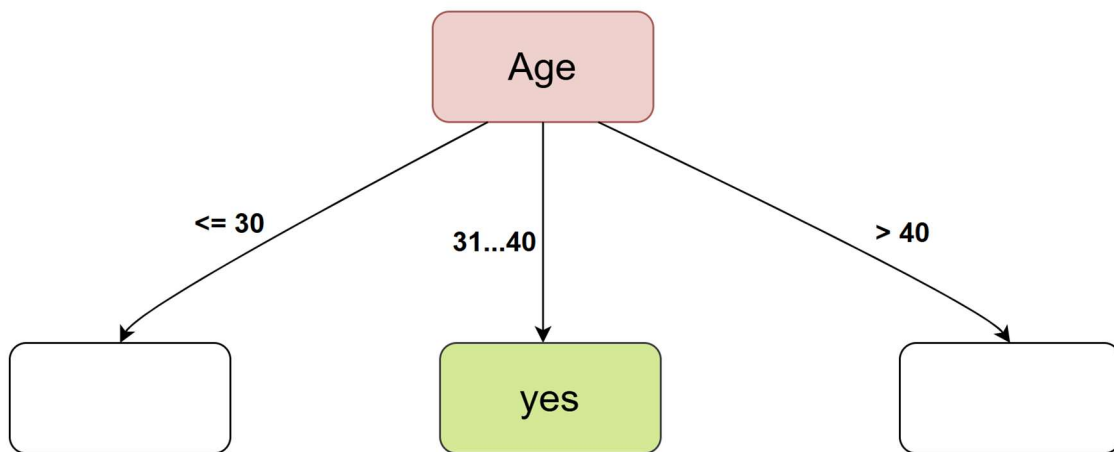
$$IG(D, \text{credit_rating}) = 0.940 - 0.892 = 0.048$$

We Know,

Root attribut = Attribute with Highest Information Gain

Root attribut = age

(4)



age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

❖ For next left side root node:

Is given,

Totals: 5 datasets $\rightarrow \underbrace{2 \text{ yes}}_{\text{class 1}}, \underbrace{3 \text{ no}}_{\text{class 2}}.$

$$\therefore N = 5$$

$$\therefore n_1 = 2$$

$$\therefore n_2 = 3$$

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{2}{5} = 0.400$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{3}{5} = 0.600$$

now,

$$Entropy(D) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(D) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$Entropy(D) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$Entropy(D) = -0.400 \log_2(0.400) - 0.600 \log_2(0.600)$$

$$Entropy(D) = 0.971$$

$\therefore$ Entropy of the entire dataset for the target attribute *buys_computer*, $D = 0.971$

❖ For Income (values: high, medium, low):

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

$$\text{Entropy}(D_{v \text{ high}}) = -\frac{2}{2} \log_2 \left(\frac{2}{2} \right) - \frac{0}{2} \log_2 \left(\frac{0}{2} \right)$$

$$\text{Entropy}(D_{v \text{ high}}) = 0.000$$

$$\text{Entropy}(D_{v \text{ medium}}) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right)$$

$$\text{Entropy}(D_{v \text{ medium}}) = 1.000$$

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

$$\text{Entropy}(D_{v \text{ low}}) = -\frac{1}{1} \log_2 \left(\frac{1}{1} \right) - \frac{0}{1} \log_2 \left(\frac{0}{1} \right)$$

$$\text{Entropy}(D_{v \text{ low}}) = 0.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ high}}|}{|D|} \text{Entropy}(D_{v \text{ high}}) + \frac{|D_{v \text{ medium}}|}{|D|} \text{Entropy}(D_{v \text{ medium}}) + \frac{|D_{v \text{ low}}|}{|D|} \text{Entropy}(D_{v \text{ low}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|2|}{|5|} \times 0.000 + \frac{|2|}{|5|} \times 1.000 + \frac{|1|}{|5|} \times 0.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.400$$

So,

We, have $\text{Entropy}(D) = 0.971$

$$\therefore \text{IG}(D, \text{Income}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Income}) = 0.971 - 0.400 = 0.571$$

❖ For Student (values: yes, no):

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

$$\text{Entropy}(D_{v \text{ no}}) = -\frac{3}{3} \log_2 \left(\frac{3}{3} \right) - \frac{0}{3} \log_2 \left(\frac{0}{3} \right)$$

$$\text{Entropy}(D_{v \text{ no}}) = 0.000$$

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

$$\text{Entropy}(D_{v \text{ yes}}) = -\frac{2}{2} \log_2 \left(\frac{2}{2} \right) - \frac{0}{2} \log_2 \left(\frac{0}{2} \right)$$

$$\text{Entropy}(D_{v \text{ yes}}) = 0.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ no}}|}{|D|} \text{Entropy}(D_{v \text{ no}}) + \frac{|D_{v \text{ yes}}|}{|D|} \text{Entropy}(D_{v \text{ yes}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|3|}{|5|} \times 0.000 + \frac{|2|}{|5|} \times 0.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.000$$

So,

We, have $\text{Entropy}(D) = 0.971$

$$\therefore \text{IG}(D, \text{Student}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Student}) = 0.971 - 0.000 = 0.971$$

❖ For credit_rating (values: fair, excellent):

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

age	income	student	credit_rating	buys
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
≤ 30	medium	yes	excellent	yes

$$\begin{aligned} \text{Entropy}(D_{v \text{ fair}}) &= -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \\ \text{Entropy}(D_{v \text{ fair}}) &= 0.918 \end{aligned}$$

$$\begin{aligned} \text{Entropy}(D_{v \text{ excellent}}) &= -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \\ \text{Entropy}(D_{v \text{ excellent}}) &= 1.000 \end{aligned}$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ fair}}|}{|D|} \text{Entropy}(D_{v \text{ fair}}) + \frac{|D_{v \text{ excellent}}|}{|D|} \text{Entropy}(D_{v \text{ excellent}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|3|}{|5|} \times 0.918 + \frac{|2|}{|5|} \times 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.951$$

So,

We, have $\text{Entropy}(D) = 0.971$

$$\therefore \text{IG}(D, \text{credit_rating}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{credit_rating}) = 0.971 - 0.951 = 0.020$$

We have,

$$\text{IG}(D, \text{Income}) = 0.940 - 0.911 = 0.571$$

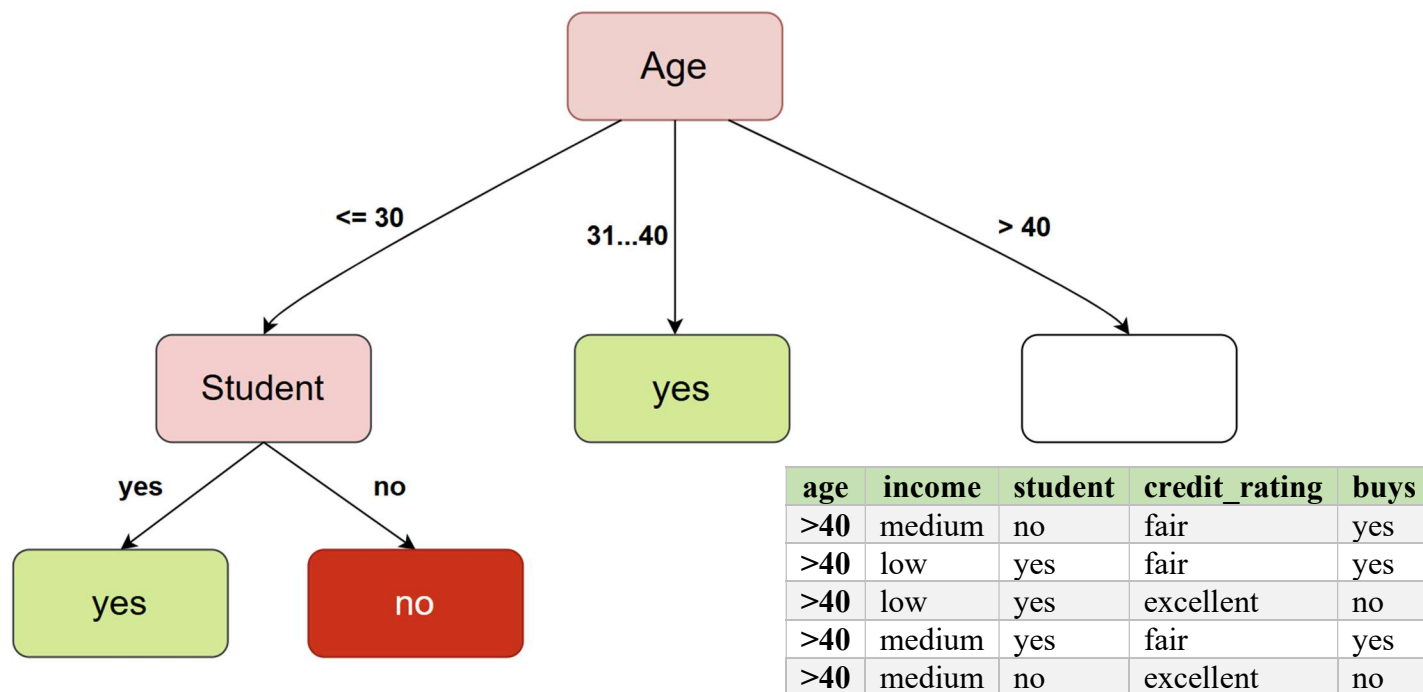
$$\text{IG}(D, \text{Student}) = 0.940 - 0.789 = 0.971$$

$$\text{IG}(D, \text{credit_rating}) = 0.940 - 0.892 = 0.020$$

We Know,

Root attribut = Attribute with Highest Information Gain

Root attribut = student



❖ For next right side root node:

Is given,

Totals: 5 datasets → $\underbrace{3}_{\text{class 1}} \text{ yes}, \underbrace{2}_{\text{class 2}} \text{ no}.$

$$\therefore N = 5$$

$$\therefore n_1 = 3$$

$$\therefore n_2 = 2$$

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{3}{5} = 0.600$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{2}{5} = 0.400$$

now,

$$Entropy(D) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(D) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$Entropy(D) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$Entropy(D) = -0.600 \log_2(0.600) - 0.400 \log_2(0.400)$$

$$Entropy(D) = 0.971$$

∴ Entropy of the entire dataset for the target attribute **buys_computer**, $D = 0.971$

❖ **For Income (values: medium, low):**

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$Entropy(D_{v \text{ medium}}) = -\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right)$$

$$Entropy(D_{v \text{ medium}}) = 0.918$$

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$Entropy(D_{v \text{ low}}) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right)$$

$$Entropy(D_{v \text{ low}}) = 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = \frac{|D_{v \text{ medium}}|}{|D|} Entropy(D_{v \text{ medium}}) + \frac{|D_{v \text{ low}}|}{|D|} Entropy(D_{v \text{ low}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = \frac{|3|}{|5|} \times 0.918 + \frac{|2|}{|5|} \times 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = 0.951$$

So,

We, have $Entropy(D) = 0.971$

$$\therefore IG(D, \text{Income}) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)$$

$$IG(D, \text{Income}) = 0.971 - 0.951 = 0.020$$

❖ For Student (values: yes, no):

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Entropy}(D_{v \text{ no}}) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right)$$

$$\text{Entropy}(D_{v \text{ no}}) = 1.000$$

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Entropy}(D_{v \text{ yes}}) = -\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right)$$

$$\text{Entropy}(D_{v \text{ yes}}) = 0.918$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ no}}|}{|D|} \text{Entropy}(D_{v \text{ no}}) + \frac{|D_{v \text{ yes}}|}{|D|} \text{Entropy}(D_{v \text{ yes}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|2|}{|5|} \times 1.000 + \frac{|3|}{|5|} \times 0.918$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.951$$

So,

We, have $\text{Entropy}(D) = 0.971$

$$\therefore \text{IG}(D, \text{Student}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Student}) = 0.971 - 0.951 = 0.020$$

❖ For credit_rating (values: fair, excellent):

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

age	income	student	credit_rating	buys
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$\begin{aligned} \text{Entropy}(D_{v \text{ fair}}) &= -\frac{3}{3} \log_2 \left(\frac{3}{3} \right) - \frac{0}{3} \log_2 \left(\frac{0}{3} \right) \\ \text{Entropy}(D_{v \text{ fair}}) &= 0.000 \end{aligned}$$

$$\begin{aligned} \text{Entropy}(D_{v \text{ excellent}}) &= -\frac{2}{2} \log_2 \left(\frac{2}{2} \right) - \frac{0}{2} \log_2 \left(\frac{0}{2} \right) \\ \text{Entropy}(D_{v \text{ excellent}}) &= 0.000 \end{aligned}$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ fair}}|}{|D|} \text{Entropy}(D_{v \text{ fair}}) + \frac{|D_{v \text{ excellent}}|}{|D|} \text{Entropy}(D_{v \text{ excellent}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|3|}{|5|} \times 0.000 + \frac{|2|}{|5|} \times 0.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.000$$

So,

We, have $\text{Entropy}(D) = 0.971$

$$\therefore \text{IG}(D, \text{credit_rating}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{credit_rating}) = 0.971 - 0.000 = 0.971$$

We have,

$$\text{IG}(D, \text{Income}) = 0.940 - 0.911 = 0.020$$

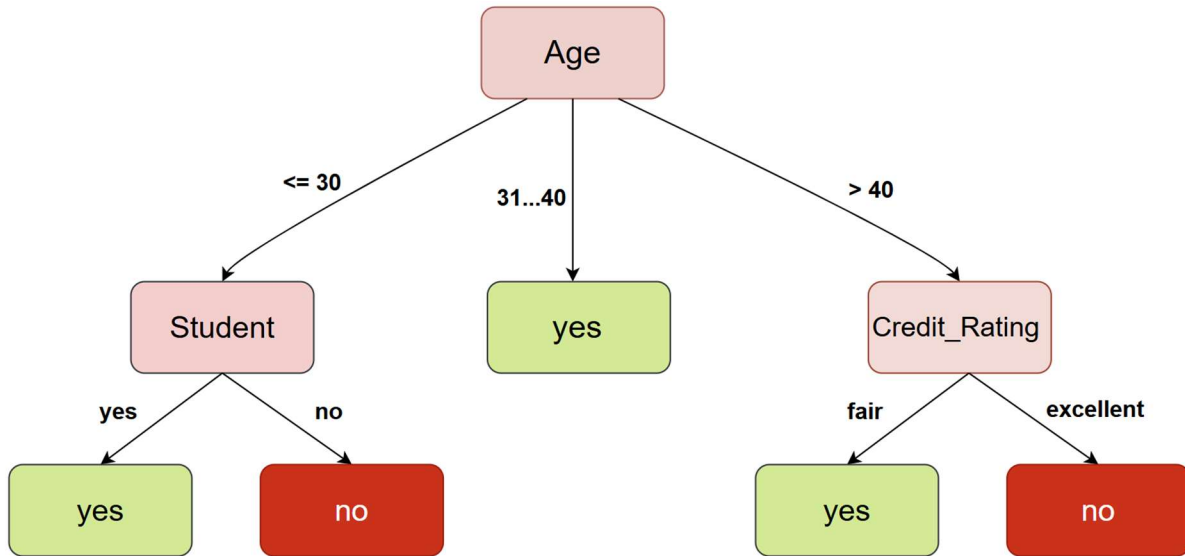
$$\text{IG}(D, \text{Student}) = 0.940 - 0.789 = 0.020$$

$$\text{IG}(D, \text{credit_rating}) = 0.940 - 0.892 = 0.971$$

We Know,

Root attribut = Attribute with Highest Information Gain

Root attribut = credit_rating

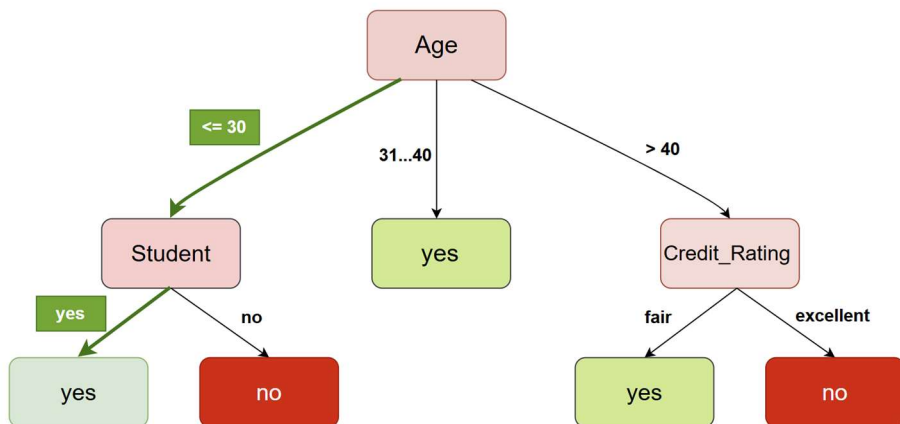


(5)

Is given,

*age = ≤ 30
 income = low
 student = yes
 credit_rating = fair*

So, Predict the value of the target attribute:



$\therefore$ buys_computer = YES

❧ **Problem2:** Consider the following dataset of 14 Records with attributes **Outlook**, **Temp**, **Humidity**, **Wind**, and the target class **Play Tennis**.

Dataset:

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

❖ **Your Tasks is:**

1. Calculate the entropy of the entire dataset for the target attribute **Play Tennis**.
2. Compute the **Information Gain (IG)** for each attribute:
 - Outlook
 - Temp
 - Humidity
 - Wind
3. Determine the best attribute for the root node based on highest Information Gain.
4. Using your root node, **perform the next split** on the corresponding subsets and show the resulting smaller decision tree.
5. Finally, use the constructed decision tree to classify the following new instance:

Outlook = Sunny

Temp = Cool

Humidity = High

Wind = Weak

Predict the value of the target attribute: **Play Tennis?**

(1)

Is given,

Totals: **14 datasets** $\rightarrow$ $\underbrace{9}_{\text{class 1}} \text{ yes}, \underbrace{5}_{\text{class 2}} \text{ no}.$

$$\therefore N = 14$$

$$\therefore n_1 = 9$$

$$\therefore n_2 = 5$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{9}{14} = 0.643$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{5}{14} = 0.357$$

now,

$$Entropy(D) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(D) = - \sum_{i=1}^2 p_i \log_2(p_i)$$

$$Entropy(D) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$Entropy(D) = -0.643 \log_2(0.643) - 0.357 \log_2(0.357)$$

$$Entropy(D) = 0.940$$

$\therefore$ Entropy of the entire dataset for the target attribute **Play Tennis**, $D = 0.940$

(2)

We Know,

$$IG(D, A) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)$$

❖ For Outlook (values: Sunny, Overcast, Rain):

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No _(sunny)
D2	Sunny	Hot	High	Strong	No _(sunny)
D3	Overcast	Hot	High	Weak	Yes _(Overcast)
D4	Rain	Mild	High	Weak	Yes _(Rain)
D5	Rain	Cool	Normal	Weak	Yes _(Rain)
D6	Rain	Cool	Normal	Strong	No _(Rain)
D7	Overcast	Cool	Normal	Strong	Yes _(Overcast)
D8	Sunny	Mild	High	Weak	No _(sunny)
D9	Sunny	Cool	Normal	Weak	Yes _(sunny)
D10	Rain	Mild	Normal	Weak	Yes _(Rain)
D11	Sunny	Mild	Normal	Strong	Yes _(sunny)
D12	Overcast	Mild	High	Strong	Yes _(Overcast)
D13	Overcast	Hot	Normal	Weak	Yes _(Overcast)
D14	Rain	Mild	High	Strong	No _(Rain)

$$\text{Entropy}(D_{v \text{ sunny}}) = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$\text{Entropy}(D_{v \text{ sunny}}) = 0.971$$

$$\text{Entropy}(D_{v \text{ overcast}}) = -\frac{4}{4} \log_2 \left(\frac{4}{4} \right) - \frac{0}{4} \log_2 \left(\frac{0}{4} \right)$$

$$\text{Entropy}(D_{v \text{ overcast}}) = 0.000$$

$$\text{Entropy}(D_{v \text{ rain}}) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right)$$

$$\text{Entropy}(D_{v \text{ rain}}) = 0.971$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ sunny}}|}{|D|} \text{Entropy}(D_{v \text{ sunny}}) + \frac{|D_{v \text{ overcast}}|}{|D|} \text{Entropy}(D_{v \text{ overcast}}) + \frac{|D_{v \text{ rain}}|}{|D|} \text{Entropy}(D_{v \text{ rain}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|5|}{|14|} \times 0.971 + \frac{|4|}{|14|} \times 0.000 + \frac{|5|}{|14|} \times 0.971$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.694$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Outlook}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Outlook}) = 0.940 - 0.694 = 0.246$$

❖ For Temp (values: Hot, Mild, Cool):

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No (Hot)
D2	Sunny	Hot	High	Strong	No (Hot)
D3	Overcast	Hot	High	Weak	Yes (Hot)
D4	Rain	Mild	High	Weak	Yes (Mild)
D5	Rain	Cool	Normal	Weak	Yes (cool)
D6	Rain	Cool	Normal	Strong	No (cool)
D7	Overcast	Cool	Normal	Strong	Yes (cool)
D8	Sunny	Mild	High	Weak	No (Mild)
D9	Sunny	Cool	Normal	Weak	Yes (cool)
D10	Rain	Mild	Normal	Weak	Yes (Mild)
D11	Sunny	Mild	Normal	Strong	Yes (Mild)
D12	Overcast	Mild	High	Strong	Yes (Mild)
D13	Overcast	Hot	Normal	Weak	Yes (Hot)
D14	Rain	Mild	High	Strong	No (Mild)

$$\begin{aligned} \text{Entropy}(D_{v \text{ Hot}}) &= -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \\ \text{Entropy}(D_{v \text{ Hot}}) &= 1.000 \end{aligned}$$

$$\begin{aligned} \text{Entropy}(D_{v \text{ Mild}}) &= -\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \\ \text{Entropy}(D_{v \text{ Mild}}) &= 0.918 \end{aligned}$$

$$\begin{aligned} \text{Entropy}(D_{v \text{ Cool}}) &= -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \\ \text{Entropy}(D_{v \text{ Cool}}) &= 0.811 \end{aligned}$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ Hot}}|}{|D|} \text{Entropy}(D_{v \text{ Hot}}) + \frac{|D_{v \text{ Mild}}|}{|D|} \text{Entropy}(D_{v \text{ Mild}}) + \frac{|D_{v \text{ Cool}}|}{|D|} \text{Entropy}(D_{v \text{ Cool}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|4|}{|14|} \times 1.000 + \frac{|6|}{|14|} \times 0.918 + \frac{|4|}{|14|} \times 0.811$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.911$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Temp}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Temp}) = 0.940 - 0.911 = 0.029$$

❖ For Humidity (values: High, Normal):

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No (High)
D2	Sunny	Hot	High	Strong	No (High)
D3	Overcast	Hot	High	Weak	Yes (High)
D4	Rain	Mild	High	Weak	Yes (High)
D5	Rain	Cool	Normal	Weak	Yes (Normal)
D6	Rain	Cool	Normal	Strong	No (Normal)
D7	Overcast	Cool	Normal	Strong	Yes (Normal)
D8	Sunny	Mild	High	Weak	No (High)
D9	Sunny	Cool	Normal	Weak	Yes (Normal)
D10	Rain	Mild	Normal	Weak	Yes (Normal)
D11	Sunny	Mild	Normal	Strong	Yes (Normal)
D12	Overcast	Mild	High	Strong	Yes (High)
D13	Overcast	Hot	Normal	Weak	Yes (Normal)
D14	Rain	Mild	High	Strong	No (High)

$$\text{Entropy}(D_{v \text{ high}}) = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right)$$

$$\text{Entropy}(D_{v \text{ high}}) = 0.985$$

$$\text{Entropy}(D_{v \text{ normal}}) = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right)$$

$$\text{Entropy}(D_{v \text{ normal}}) = 0.592$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|D_{v \text{ High}}|}{|D|} \text{Entropy}(D_{v \text{ High}}) + \frac{|D_{v \text{ Normal}}|}{|D|} \text{Entropy}(D_{v \text{ Normal}})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = \frac{|7|}{|14|} \times 0.985 + \frac{|7|}{|14|} \times 0.592$$

$$\therefore \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v) = 0.789$$

So,

We, have $\text{Entropy}(D) = 0.940$

$$\therefore \text{IG}(D, \text{Humidity}) = \text{Entropy}(D) - \sum \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{IG}(D, \text{Humidity}) = 0.940 - 0.789 = 0.151$$

❖ For Wind (values: Weak, Strong):

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No (Weak)
D2	Sunny	Hot	High	Strong	No (Strong)
D3	Overcast	Hot	High	Weak	Yes (Weak)
D4	Rain	Mild	High	Weak	Yes (Weak)
D5	Rain	Cool	Normal	Weak	Yes (Weak)
D6	Rain	Cool	Normal	Strong	No (Strong)
D7	Overcast	Cool	Normal	Strong	Yes (Strong)
D8	Sunny	Mild	High	Weak	No (Weak)
D9	Sunny	Cool	Normal	Weak	Yes (Weak)
D10	Rain	Mild	Normal	Weak	Yes (Weak)
D11	Sunny	Mild	Normal	Strong	Yes (Strong)
D12	Overcast	Mild	High	Strong	Yes (Strong)
D13	Overcast	Hot	Normal	Weak	Yes (Weak)
D14	Rain	Mild	High	Strong	No (Strong)

$$Entropy(D_{v Weak}) = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right)$$

$$Entropy(D_{v Weak}) = 0.811$$

$$Entropy(D_{v Strong}) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right)$$

$$Entropy(D_{v Strong}) = 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = \frac{|D_{v Weak}|}{|D|} Entropy(D_{v Weak}) + \frac{|D_{v Strong}|}{|D|} Entropy(D_{v Strong})$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = \frac{8}{14} \times 0.811 + \frac{6}{14} \times 1.000$$

$$\therefore \sum \frac{|D_v|}{|D|} \times Entropy(D_v) = 0.892$$

So,

We, have $Entropy(D) = 0.940$

$$\therefore IG(D, Wind) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)$$

$$IG(D, Wind) = 0.940 - 0.892 = 0.048$$

(3)

We have,

$$IG(D, \text{Outlook}) = 0.940 - 0.694 = 0.246$$

$$IG(D, \text{Temp}) = 0.940 - 0.911 = 0.029$$

$$IG(D, \text{Humidity}) = 0.940 - 0.789 = 0.151$$

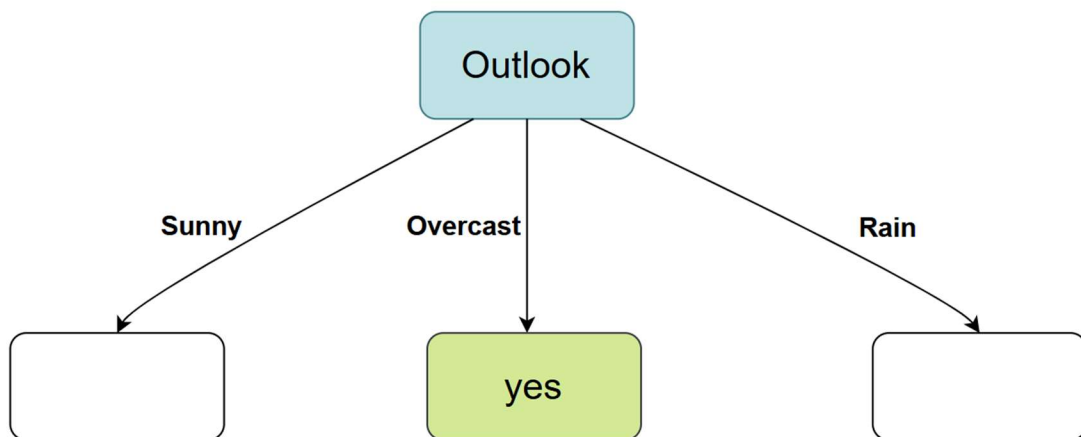
$$IG(D, \text{Wind}) = 0.940 - 0.892 = 0.048$$

We Know,

Root attribut = Attribute with Highest Information Gain

Root attribut = Outlook

(4)

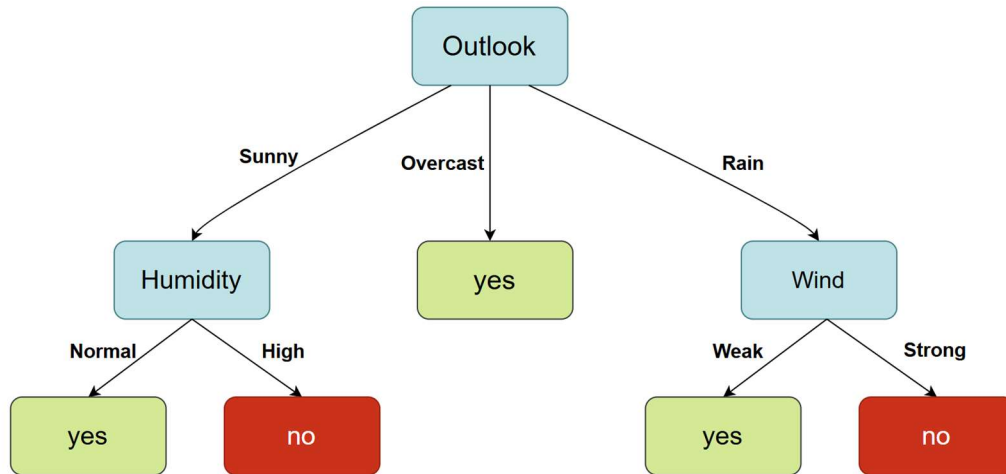


Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Outlook	Temp	Humidity	Wind	Play Tennis
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Next, repeat the same process with the other nodes with smaller dataset

FINAL DECISION TREE

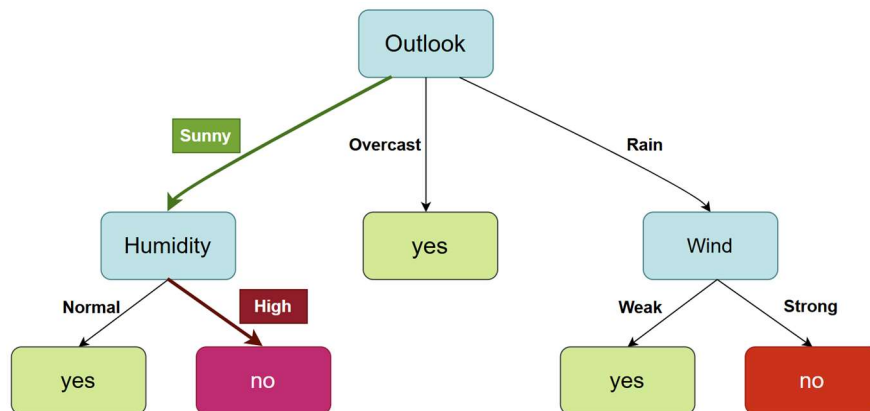


(5)

Is given,

*Outlook = Sunny
Temp = Cool
Humidity = High
Wind = Weak*

So, Predict the value of the target attribute:



∴ Play Tennis = NO

Gain Ratio

- ⇒ Gain Ratio is an **attribute selection measure** used in the **C4.5 Decision Tree algorithm**.
- ⇒ It is an improvement over **Information Gain (IG)**.

Why do we need Gain Ratio?

- ⇒ Information Gain sometimes has a **bias**:
 - It favors attributes with **many distinct values**
 - Example: A unique ID attribute (like Customer ID) would split each row separately, giving very high IG — but it's **useless** for prediction.
- ⇒ To fix this problem, **C4.5** uses **Gain Ratio**, which adjusts IG by considering how broadly an attribute splits the data.

Formula for Gain Ratio

$$\text{GainRatio}(A) = \frac{\text{I. Gain}(A)}{\text{SplitInfo}(A)}$$

Where:

$$\text{I. Gain}(A) = \text{Entropy}(D) - \sum_{i=1}^v \frac{|D_v|}{|D|} \times \text{Entropy}(D_v)$$

$$\text{SplitInfo}(A) = - \sum_{i=1}^v \frac{|D_v|}{|D|} \times \log_2 \left(\frac{|D_v|}{|D|} \right)$$

🔗 **Problem3:** Compute the **Gain Ratio** for the attribute **Outlook** using the Play Tennis dataset. Provide the Information Gain, SplitInfo, and the final Gain Ratio.

We have,

$$IG(D, \text{Outlook}) = 0.940 - 0.694 = 0.246$$

We Know,

Outlook	Count	Proportion (p)
Sunny	5	5/14 = 0.3571
Overcast	4	4/14 = 0.2857
Rain	5	5/14 = 0.3571

$$\text{SplitInfo}(A) = - \sum_{i=1}^v \frac{|D_v|}{|D|} \times \log_2 \left(\frac{|D_v|}{|D|} \right)$$

$$\text{SplitInfo}(\text{Outlook}) = -\frac{5}{14} \log_2 \frac{5}{14} - \frac{4}{14} \log_2 \frac{4}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$\text{SplitInfo}(\text{Outlook}) = 1.577$$

So,

$$\text{GainRatio}(\text{Outlook}) = \frac{IG(A)}{\text{SplitInfo}(A)}$$

$$\text{GainRatio}(\text{Outlook}) = \frac{0.246}{1.577} = 0.156$$

Gini Index

⇒ The **Gini Index** (or Gini Impurity) is a measure used in **CART (Classification and Regression Trees)** to select the best attribute for splitting data.

→ It measures **how impure** a dataset is.

- **Gini = 0** → perfectly pure node (all samples belong to one class)
- **Higher Gini** → more mixed classes (more impurity)

Why use Gini Index?

- Faster to compute than entropy (no logarithms)
- Works well in practice
- Does **not** have bias toward attributes with many values (unlike Information Gain)
- Used in CART, Random Forest, and modern tree-based models

Gini Index Formula

$$\text{Gini}(S) = 1 - \sum_{i=1}^c (p_i)^2$$

Where:

- p_i = proportion of class i
- c = number of classes

$$\text{SplitInfo}(S) = \sum_{i=1}^v \frac{|S_v|}{|S|} \text{Gini}(S_v)$$

🔗 **Problem4:** Using the Play Tennis dataset below, compute the **Gini Index** to determine the best attribute for splitting the dataset. Show all steps clearly.

Dataset:

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Your Tasks is:

- Compute the **Gini Index** of the entire dataset.
- Compute the **Gini Index** for each attribute (Outlook, Temp, Humidity, Wind).
- Determine which attribute provides the **best split** according to Gini Index.
- Show the weighted Gini Index calculation for the chosen attribute.

(a)

Is given,

Totals: **14 datasets** → $\underbrace{9}_{\text{class 1}} \text{ yes}, \underbrace{5}_{\text{class 2}} \text{ no}.$

$$\therefore N = 14$$

$$\therefore n_1 = 9$$

$$\therefore n_2 = 5$$

We Know,

$$\text{proportion of class } i = p_i = \frac{n_i}{N}$$

$$\text{proportion of class 1} = p_1 = \frac{n_1}{N} = \frac{9}{14} = 0.643$$

$$\text{proportion of class 2} = p_2 = \frac{n_2}{N} = \frac{5}{14} = 0.357$$

Now,

$$\text{Gini}(S) = 1 - \sum_{i=1}^c (p_i)^2$$

$$\text{Gini}(S) = 1 - \sum_{i=1}^2 (p_i)^2$$

$$\text{Gini}(S) = 1 - \{p_1^2 + p_2^2\}$$

$$\text{Gini}(S) = 1 - \{(0.643)^2 + (0.357)^2\}$$

$$\text{Gini}(S) = 0.459$$

(b)

❖ **Compute Gini Index for a Split for Outlook:**

⇒ Outlook has three values:

Outlook	Yes	No	Total
Sunny	2	3	5
Overcast	4	0	4
Rain	3	2	5

⇒ Gini for each branch:

$$\text{Gini}(\text{Sunny}) = 1 - \left\{ \left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right\} = 0.480$$

$$\text{Gini}(\text{Overcast}) = 1 - \left\{ \left(\frac{4}{4} \right)^2 + \left(\frac{0}{4} \right)^2 \right\} = 0.000$$

$$\text{Gini}(\text{Rain}) = 1 - \left\{ \left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right\} = 0.480$$

⇒ Weighted Gini Index for Outlook:

$$\text{Gini}_{\text{split}} = \frac{5}{14}(0.48) + \frac{4}{14}(0) + \frac{5}{14}(0.48) = 0.343$$

❖ Compute Gini Index for a Split for TEMP:

⇒ TEMP has three values:

Temp	Yes	No	Total
Hot	2	2	4
Mild	4	2	6
Cool	3	1	4

⇒ Gini for each branch:

$$\text{Gini(Hot)} = 1 - \left\{ \left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right\} = 0.500$$

$$\text{Gini(Mild)} = 1 - \left\{ \left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right\} = 0.444$$

$$\text{Gini(Cool)} = 1 - \left\{ \left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right\} = 0.375$$

⇒ Weighted Gini Index for Temp:

$$\text{Gini}_{\text{split}} = \frac{4}{14} (0.500) + \frac{6}{14} (0.444) + \frac{4}{14} (0.375) = 0.440$$

❖ Compute Gini Index for a Split for Humidity:

⇒ Humidity has two values:

Humidity	Yes	No	Total
High	3	4	7
Normal	6	1	7

⇒ Gini for each branch:

$$\text{Gini(High)} = 1 - \left\{ \left(\frac{3}{7} \right)^2 + \left(\frac{4}{7} \right)^2 \right\} = 0.490$$

$$\text{Gini(Normal)} = 1 - \left\{ \left(\frac{6}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right\} = 0.245$$

⇒ Weighted Gini Index for Humidity:

$$\text{Gini}_{\text{split}} = \frac{7}{14} (0.490) + \frac{7}{14} (0.245) = 0.368$$

❖ Compute Gini Index for a Split for Wind:

⇒ Wind has two values:

Wind	Yes	No	Total
Weak	6	2	8
Strong	3	3	6

⇒ Gini for each branch:

$$\text{Gini(Weak)} = 1 - \left\{ \left(\frac{6}{8} \right)^2 + \left(\frac{2}{8} \right)^2 \right\} = 0.375$$

$$\text{Gini(Strong)} = 1 - \left\{ \left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right\} = 0.500$$

⇒ Weighted Gini Index for Wind:

$$\text{Gini}_{\text{split}} = \frac{7}{14}(0.490) + \frac{7}{14}(0.245) = 0.368$$

(c)

We have,

Attribute	Gini Split
Outlook	0.343
Temperature	0.441
Humidity	0.367
Wind	0.428

We Know,

∴ BEST ATTRIBUTE = OUTLOOK

Because it has the **lowest weighted Gini Index**

Thus, **CART Decision Tree will split on Outlook first.**

(d)

⇒ Weighted Gini Index for Outlook:

$$\text{Gini}_{\text{split}} = \frac{5}{14}(0.48) + \frac{4}{14}(0) + \frac{5}{14}(0.48) = 0.343$$

Overfitting in Decision Trees

❖ What is it?

- Tree becomes **too complex**.
- Fits noise instead of patterns.
- Symptoms:
 - High training accuracy.
 - Low test accuracy.

❖ Causes

- Tree grown too deep.
- Too many branches capturing random noise.

❖ Result

- Poor generalization.

Pre-pruning (Early Stopping)

⇒ Stop tree growth **before it becomes too complex**.

❖ Stopping criteria

- Max tree depth reached.
- Minimum samples at leaf.
- Minimum info gain threshold.

❖ Example

- If $IG < 0.01 \rightarrow$ stop splitting.

❖ Advantages

- Prevents overfitting
- Faster training
- Simple, interpretable tree

❖ Disadvantages

- May stop early and miss good splits
- Hard to choose thresholds

Post-Pruning (Cost Complexity Pruning)

⇒ First grow **full tree**, then prune unnecessary branches.

❖ Steps

1. Grow full tree.
2. Starting from leaves, remove branches that don't improve accuracy.
3. Use validation set or cross-validation.

❖ Example

- In spam classification, deep branches with rare word patterns are pruned.

❖ Advantages

- Better generalization
- Simplifies tree
- Data-driven pruning

❖ Disadvantages

- Extra computation
- Needs validation set
- Time-consuming on large trees